

Forman Christian College (A Chartered University)
Embedded Systems (CSCS306)
Assignment 1
4 Digit Stop Watch Using Arduino

Group of two persons is allowed in this assignment. You can discuss the working of assignment with your class fellows. Sharing any part of your assignment is strictly prohibited. Any such attempt will result in zero marks in this assignment.

Date of Submission: upload your code file and soft copy of report on Moodle course page on or before Monday Oct 14 11:59 pm. The working hardware should be presented on Tuesday Oct 15, 2024 between 11:00 and 11:20 am in class. Any submission after the specified date and time will be considered late. A penalty of 50% marks deduction per day will be applicable on late submission.

Grading Criteria

Working Software interfaced with the hardware: 70%

Report: 20%

Neat wiring: 10% (Loops of jumper wires that clutter the display will not earn you this 20%)

For neat wiring use twisted pair data wires.

Assignment Task

In this assignment you need to design and make a stop watch. This stop watch should have four multiplexed 7-segment displays. The 7-segment module should be capable of counting from 0 to 99 (up count as well as down count) on the two most significant digits. The two least significant digits are used to display seconds, counting 01 to 60.

You also need to interface three push-to ON switches. You may use a keypad for this purpose. First switch should adjust units digit while second switch should adjust the tens digit of the two most significant ones. This means if we want to set a value of 25 on the module, we will use units switch to place 5 and tens switch to place 2 on the module. The seconds section should display 00.

Once time is adjusted, user should press third switch and this should start the stop watch. The stop watch decrements its numeric value with each minute while the least significant section should display the count down for seconds.

When time's up, all four digits should display FCCU and blink the text for three to four times before our stopwatch initializes back to all zeros.

You can get the code from Internet or may ask ChatGPT to help you. Remember what ever you submit for this assignment, you will stand responsible for this and we may call you for a viva any time after the submission date.

You may face problem with the available number of pins on Arduino. Explore Internet to find a way around this.

You should submit the working hardware on the given date and time, along with a neat and properly formatted report. Make sure you use Times New Roman in your document with appropriate font size for text, heading and sub headings. Code must have Courier font style.

Your report should carry following sections:

1. A title page displaying
 - FCC logo
 - Assignment number
 - Assignment title
 - Group members names and roll numbers.
2. An Introduction page that should have
 - Brief introduction about the assignment.
 - Components used
3. Circuit diagram. Hand drawn diagram will not get full marks. Try to use a software like Fritzing.

4. Image of the working hardware
5. Code
 - Proper indentation and comments are required.
6. Assignment handout
7. References